

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1842

Juneau 100 kV
June 16, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1842. The Interconnection Customer proposes a 212.7 MW wind facility interconnecting at the Juneau 100 kV substation in the Ohio area. The request was received on July 31, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 679 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$30,911,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Juneau-Reasnor 100.0 kV
Contingency: Juneau-Buntzen No. 1 100.0 kV
Loading: 130.6% (Rating: 149 MVA)
DFAX: 50.9%
Pre-project loading: 57.9%
2. Facility: Reasnor-Grnet 100.0 kV
Contingency: Buntzen No. 1-Juneau 100.0 kV L
Loading: 123.8% (Rating: 146 MVA)
DFAX: 50.9%
Pre-project loading: 49.6%
3. Facility: Goldtown-Garnet 100.0 kV
Contingency: Buntzen No. 1-Juneau 100.0 kV
Loading: 121.7% (Rating: 142 MVA)
DFAX: 50.9%
Pre-project loading: 45.6%
4. Facility: Sonora-Frick Green Ridge Power 100.0 kV
Contingency: Magnolia Ranch N-Lick 100.0 kV
Loading: 117.7% (Rating: 118 MVA)
DFAX: 26.1%
Pre-project loading: 70.6%
5. Facility: Buntzen No. 1-Juneau 100.0 kV Line
Contingency: Reasnor-Juneau 100.0 kV
Loading: 117.3% (Rating: 141 MVA)
DFAX: 49.6%
Pre-project loading: 42.2%

6. Facility: Buntzen No. 1-Humboldt 100.0 kV Line
Contingency: Reasnor-Juneau 100.0 kV
Loading: 105.8% (Rating: 146 MVA)
DFAX: 49.6%
Pre-project loading: 33.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Juneau-Reasnor 100.0 kV	Rebuild line with higher-capacity conductor	\$6,696,000	73% (\$4,915,000)
Garnet-Reasnor 100.0 kV line	Rebuild line with higher-capacity conductor	\$9,934,000	73% (\$7,291,000)
Garnet-Goldtown 100.0 kV Line	Rebuild line with higher-capacity conductor	\$4,617,000	73% (\$3,389,000)
Sonora-Frick Green Ridge Power 100.0 kV	Rebuild line with higher-capacity conductor	\$10,977,000	27% (\$2,946,000)
Buntzen No. 1-Juneau 100.0 kV Line	Rebuild line with higher-capacity conductor	\$15,877,000	73% (\$11,647,000)
Humboldt-Buntzen No. 1 100.0 kV Line	Reconductor line	\$985,000	73% (\$723,000)
	Total allocated to project		\$30,911,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.